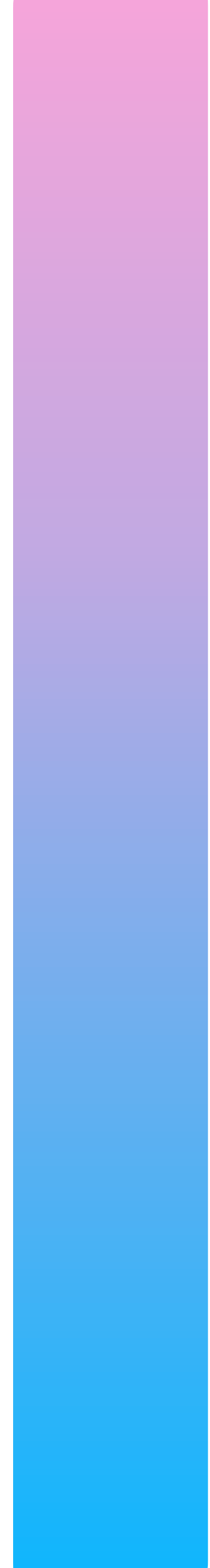


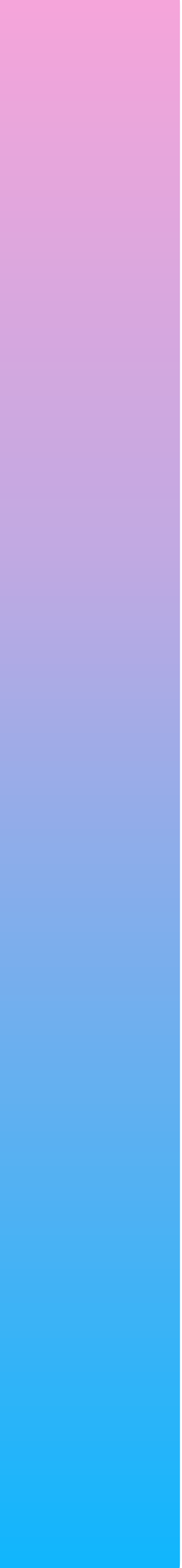
POLIMEME & MEMPERIL

DATASETS



POLIMEME

Exploring offensive meme propagation in the Israel-Palestine conflict



POLIMEME



- Multimodal dataset encompassing text-embedded memes relevant to the Israel-Palestine conflict, annotated for sentiment analysis (positive, negative, neutral) and offensive content detection, further subdivided into **direction** of offense (directed vs. undirected) and **targeted entities** (individuals, organizations, communities).

DATA COLLECTION & ANALYSIS



Data Collection

- **2,828** text-embedded images collected from diverse platforms, including **Reddit**, **Facebook**, **Instagram**, and **Google Search**
- Chosen based on their clear **association with the conflict** and their potential to convey **strong sentiments** or offensive speech.
- Manually annotated images to identify **Sentiment**, **Offensive** speech, the **direction** of offense, and the **targets** of directed offensive speech.

Data Analysis

- Preliminary analysis of the dataset to establish benchmarks using various textual, visual, and multimodal algorithms
- Experiments demonstrate that utilizing multiple modalities is crucial for a more accurate understanding of offensive speech in text-embedded images

DATA COLLECTION



Methods used for data collection

- **Reddit:** Python's PRAW (**P**ython **R**eddit **A**PI **W**rapper) to scrape text-embedded images from relevant subreddits based on specific keywords and phrases associated with the conflict
- **Google Search: Web scraping** using **BeautifulSoup** and **Selenium** to automate keyword searches for memes. BeautifulSoup handled webpage parsing, while Selenium managed dynamic loading and interactions.
- **Instagram and Facebook:** Due to **restrictions** on automated scraping, they **manually** collected memes by identifying hashtags and user account

EXAMPLES



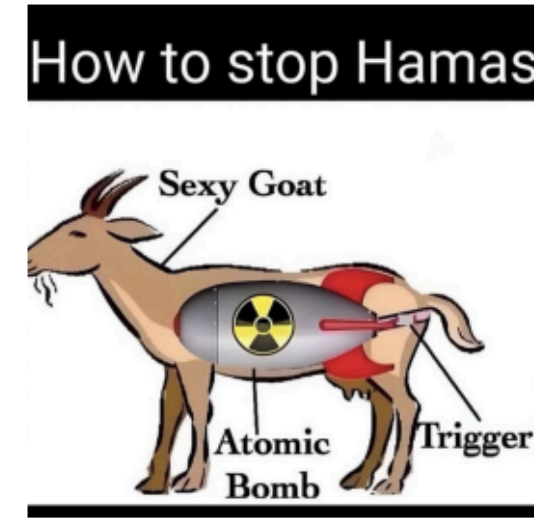
(a) Non-offensive



(b) Offensive



(c) Tar: Individual



(d) Tar: Organization



(e) Tar: Community

Figure 1: Representative text-embedded images across Non-Offensive and Offensive categories. Images for Directed Offensive content were further annotated for target classes



(a) Positive



(b) Neutral



(c) Negative

Figure 2: Example images for sentiment classification

KEY FINDINGS



- Despite their widespread use and influence, **memes often evade the level of scrutiny applied to traditional media**
- Memes can serve as potent instruments for **shaping public opinion**, including the dissemination of misinformation, offensive content, and speech. They are especially influential in political contexts, offering a straightforward yet powerful means of disseminating partisan view points and influencing public discourse
- Increasing need to identify and moderate offensive and harmful

RELATED WORK

- **Facebook Hateful Memes Challenge dataset introduced**

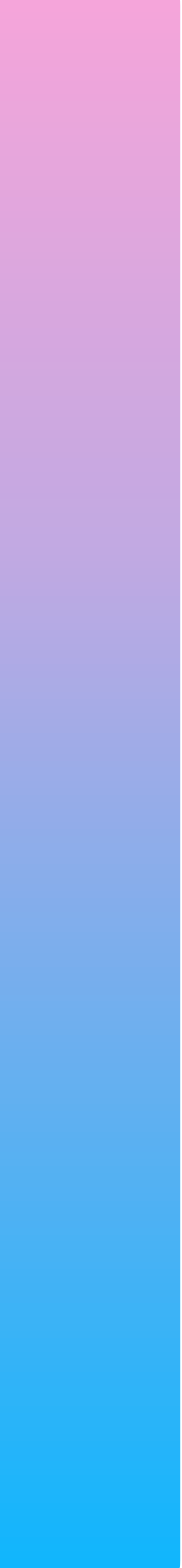
Synthetic image-text meme pairs designed to push models toward genuine multimodal understanding by conveying opposite meanings through the two modalities. Each meme is annotated for offensive content, ensuring that effective classification requires joint analysis of textual and visual cues rather than relying solely on one modality

- **Memotion**

While not exclusively focused on offense, it includes rich annotations regarding offensive content, sentiment, and intensities of emotion.

MEMEPERIL

A diverse multi-modal harmful memes dataset for evolution analytics
and LLM prompts



MEMEPERIL



- Diverse, multi-lingual, multi-modal, and temporal dataset of memes from subreddits. Focused on memes that spread harmful content, such as hate speech and bullying
- It provides a valuable insight to classifiers with user-engagement and community sentiments for memes

DATA COLLECTION

Data type

- **20,416 meme posts** and **18,279 comments** from diverse sources and multimodal content, covering text, image, and language variations
- Temporal metadata and evolution features, including repost **timestamps**, **engagement trajectory**, and **diffusion depth**, enabling research on virality
- **Human-annotated** reasoning phrases, **explaining why memes are harmful**, supporting interpretability and cognitive analysis.

Table 2 Topics and Subreddits in MemePeril dataset	
Topics	Subreddits
Political India	PoliticalHumor, DankIndia, IndiaMeme
Science	PhysicsMemes, MythologyMemes, HistoryMemes
Operating Systems	DiscordMemes, EdgeLordMemes
Religious	ReligiousFruitCake Memes
Sports, Pop Culture and other trendy topics	EdgeLordMemes, Memes_Of_TheDank, Terrible-FacebookMemes, RareInstults, DankeExchange, DankMeme2, Memes, Meme, MoldyMemes, HoIUp, FailGags, DiscordMemes, NotLikeTheO-therGirls, Funny, FingMemes

Methods used for data collection

- **Reddit:** Python’s PRAW (**Python Reddit API Wrapper**) to gather meme-related posts and comments. PRAW allows access for developers to extract posts, comments, and users
- On Reddit, posts containing strong harm are often banned, therefore, **manual filtering** was done during scraping to keep the distribution of offensive content slightly higher

DATA ANALYSIS



Data Analysis

- Analyses the harmfulness of a meme post based on the **number of likes or comments**
- The pre-trained **BERT** (bert-base-uncased) 2 model is used to **predict whether the comments posted for a meme are positive, neutral, or negative** to calculate the percentage of positive, neutral, and negative comments for each post.

MEMEPERIL FEATURES

Table 4 Count of 3 labels for Image+Text, Image, and Text

Image+Text	Image	Text	Count
Non Harmful	Non Harmful	Non Harmful	2063
Harmful	Non Harmful	Harmful	434
Harmful	Harmful	Harmful	216
Harmful	Harmful	Non Harmful	143
Harmful	Non Harmful	Non Harmful	118
Severely Harmful	Non Harmful	Severely Harmful	38
Severely Harmful	Severely Harmful	Non Harmful	20
Severely Harmful	Severely Harmful	Severely Harmful	15
Severely Harmful	Harmful	Harmful	13
Severely Harmful	Harmful	Non Harmful	12
Severely Harmful	Harmful	Severely Harmful	10
Severely Harmful	Non Harmful	Non Harmful	7
Severely Harmful	Non Harmful	Harmful	7
Harmful	Severely Harmful	Non Harmful	6
Severely Harmful	Severely Harmful	Harmful	6
Harmful	Non Harmful	Severely Harmful	5
Harmful	Severely Harmful	Harmful	4
Harmful	Harmful	Severely Harmful	1
Harmful	Severely Harmful	Severely Harmful	1

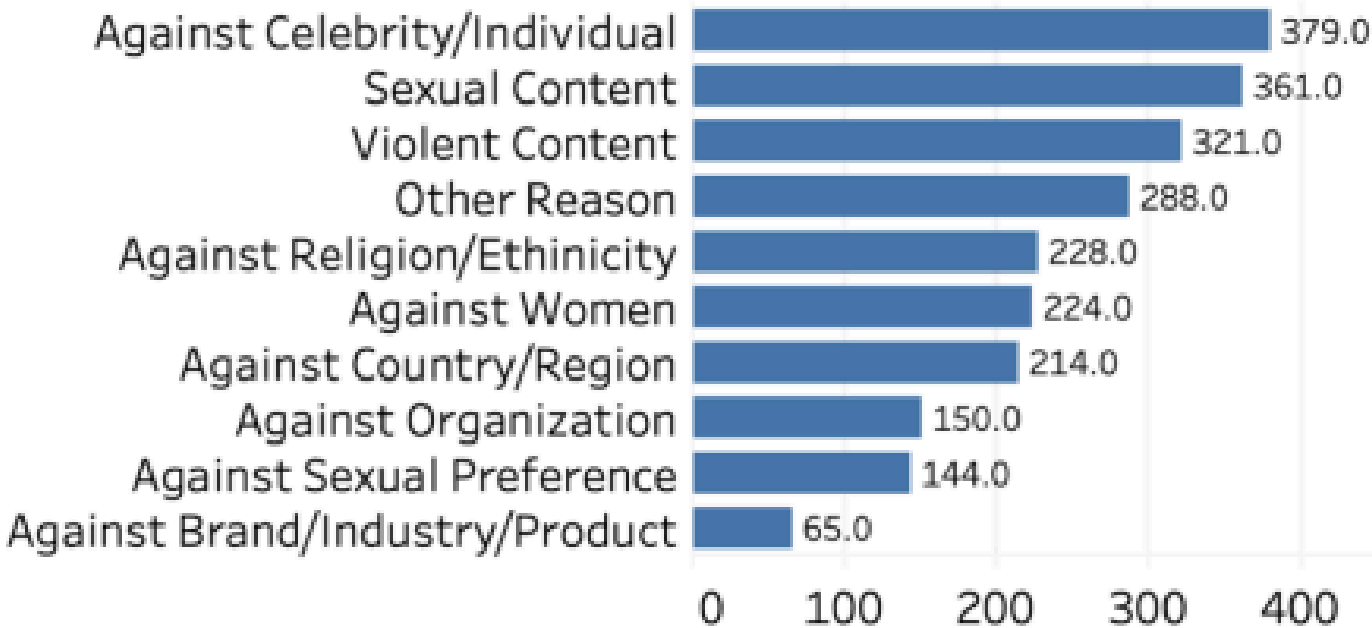


Fig. 4 Count Distribution of 'Reasons for Harmfulness'

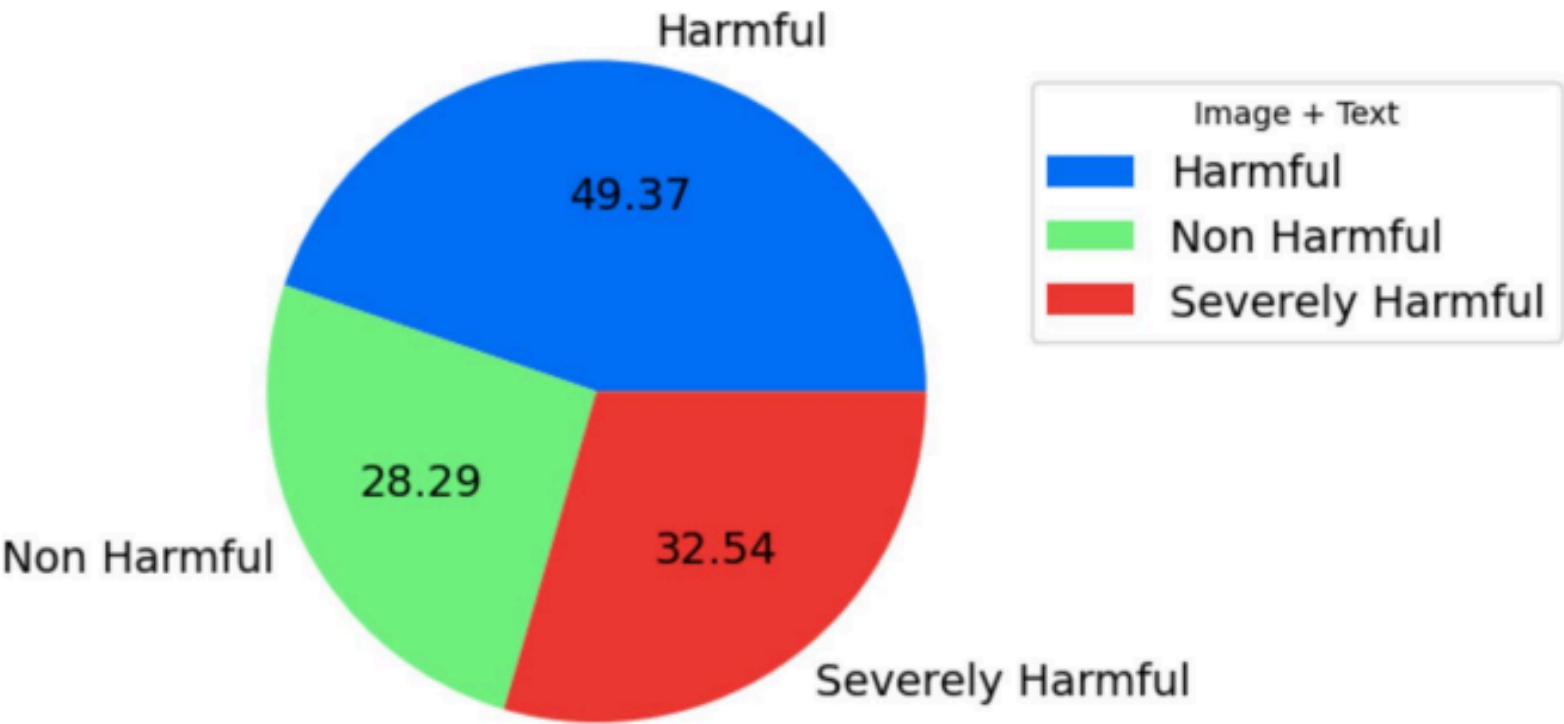


Fig. 5 Average number of comments distribution

RELATED
WORK

Table 1 Literature Survey on Dataset Type, Features, Models, Country, Technical Utility for Research studies

Research studies	Hate Lingo [20]	Hindi English Code Mix [21]	Dis-arm [22]	Mom-enta [23]	Cyberbully Classifica-tion [24]	Multi-modal Memes [25]	Our Meme Peril
Dataset Type							
Text	✓	✓	✓	✓	✓	✓	✓
Image	✗	✗	✓	✓	✗	✗	✓
Language	✗	✓	✓	✓	✗	✗	✓
Combined Text and Language	✗	✓	✗	✗	✓	✓	✓
Dataset Features							
Local Perspectives	✗	✗	✗	✗	✓	✗	✓
Cyberbullying metadata	✗	✗	✗	✗	✓	✗	✓
Evolution	✗	✗	✗	✗	✓	✗	✓
Reason Annotations / Harm Explanation	✗	✗	✗	✗	✓	✗	✓
Models Used							
Linguistic	✓	✗	✗	✗	✗	✗	✓
Character and Word N-grams	✗	✓	✗	✗	✗	✗	✗
BERT and Res-Net	✗	✗	✓	✓	✗	✗	✓
Country							
U.S	✓	✗	✗	✗	✓	✓	✗
India	✗	✓	✗	✗	✗	✗	✓
Technical Utility							
Manual Annotation	✓	✓	✓	✓	✓	✓	✓
Multi-Language Support	✗	✓	✓	✓	✗	✗	✓
Web Scraping	✗	✗	✓	✓	✗	✓	✓
Twitter API	✓	✓	✗	✗	✓	✗	✗
Google Knowledge Graph-API	✗	✗	✗	✓	✗	✗	✗

✓ indicates the presence of the feature in that study

MemePeril uniquely combines temporal metadata, human annotations, multimodal, and multilingual content, distinguishing it from previous datasets.

KEY TAKEAWAYS



What We Learned

- Meme datasets model meaning using fixed labels
- These labels assume a single ground truth

Limitations

- Meme interpretation is context-dependent and subjective
- Datasets do not capture this variability
- No modeling of differences across users or communities
- Ambiguity is reduced to single categories

Research Opportunity

- Move from fixed labels to interpretation as a distribution
- Compare dataset labels vs human perception
- Analyze divergence across groups
- Connect meaning with engagement and context

A vertical bar on the left side of the image with a gradient from pink at the top to blue at the bottom.

THANK YOU

